

Ultrafast Beam Steering User Localization in Optical Wireless Communication

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Abstract: An ultrafast user localization scheme in beam steering optical wireless communication based on real-time Fourier Transform is demonstrated. All-optical dispersive operation enables ultrafast and real-time user-specific wavelength identification overcoming bottleneck in digital signal processing. © 2021

1. Introduction

With the development of radio wireless communication, it has hit the limits of the available radio spectrum. Optical wireless communication (OWC) has been widely used in indoor high-speed communications due to the advantage of license-free optical spectrum and broad bandwidth available [1]. OWC in conjunction with beam steering has become an emerging area of interest to transmit information to different users precisely in an indoor environment [2]. A key challenge in indoor beam forming OWC is to track user terminals with high mobility in real time. In a wavelength-controlled beam steering OWC system [3], ultrafast user localization has been achieved by precisely identification instantaneous microwave frequency associated with a particular optical wavelength. As most existing work on user localization, the work in [3] relies on the detected waveform being sampled and processed using digital signal processing (DSP) approaches. It is therefore highly demanded to achieve real-time data processing in the optical domain directly. On the other hand, the optical dispersive Fourier transform (FT) approach has been developed to identify instantaneous microwave frequency in an all-optical manner [4, 5], which avoids electronic and digital bottleneck and greatly reduces energy consumption [6].

In this paper, an all-optical dispersive Fourier transform approach has been explored, for the first time to the best of knowledge, in real-time user localization in wavelength-controlled beam steering OWC system. A 45° tilted fibre grating (45° TFG) is used as a high-efficiency in-fibre diffraction device for beam steering and at the same time collecting the echo optical pulses from remote users. Different from the digital short-time Fourier transform approach used in previous systems, the reflected optical pulses are compressed by opposite dispersion to achieve real-time dispersive Fourier transform in the optical domain. After optical pulse compression, the carrier microwave frequency of reflected pulses can be determined from the time difference between splitted two pulses corresponding to the two sidebands of microwave signal. Hence, with an in-fibre beam steering structure and instantaneous microwave frequency identification via optical FT, the position of users can be decided.

2. Principle

The schematic of the proposed ultrafast user localization system is shown in Fig. 1. The proposed indoor user localization system can be divided into these parts, ultrashort pulse time-stretching, chirp coding, beam steering, and data processing.

An ultrashort pulse is temporally stretched, and then chirp encoded by an unbalanced Mach-Zehnder interferometric structure [3]. After that, the mapping between frequencies and time-domain is established. Then the 45° TFG is used to achieve ultrafast wavelength controlled beam steering. According to the characteristics of beam steering, different wavelength with different microwave frequencies is dispatched to different users. Hence, the chirped signal is steered to different users according to the frequency-time mapping. The reflected pulses from remote users will be collected by the same 45° TFG and send to a dispersion compensating fibre (DCF), where the reflected pulses will be compressed due to opposite dispersion. The compressed optical pulse will be splitted into two pulses after dispersive Fourier transform, corresponding to the two sidebands of microwave frequency. The splitted pulses are detected at a photodetector (PD) and measured by a real-time oscilloscope (OSC). We can get the central microwave frequency of the reflected pulses by measuring the time delay difference between the splitted two pulses.

The proposed ultrafast user localization scheme based on optical dispersive Fourier transform is verified by numerical simulations in this work. In the simulation, a passive mode-locked laser (MLL) is used as an optical source generates a 50-MHz optical pulse train. The emitted ultrashort pulses are time-stretched from pulse width of

550 fs to 12 ns by using optical single mode fibre of 25 km, which can achieve the frequency-to-time mapping. An unbalanced MZI structure [7] is employed to encode the pulse spectrum and get a high chirp rate due to frequency-to-time mapping. Microwave frequency chirp from ~ 7 GHz to ~ 29 GHz within the stretched pulse window is obtained. Then the pulse with a high chirp rate is steered from a 45° TFG to indoor users. During the beam steering, the frequency-time-space mapping is achieved. Each user receives an optical pulse with a different central frequency within the channel bandwidth (~ 22 GHz). After that, the reflected pulses are collected by the same 45° TFG, and are compressed by a dispersion compensating fibre (DCF) with a total dispersion value of $\tilde{\Phi}_2$. The frequency of the unknown optical pulse is determined by linear mapping relations on the distance between the adjacent pulse on OSC. Fig. 2(a) shows the linear mapping relations between the frequency and the time difference in the free-space, which has a value of ~ 20 ps/GHz. Four active users with different distances (0.4 m, 0.6 m, 1 m, and 1.5 m) are set in this simulation. The central frequencies are 11 GHz, 14.5 GHz, 15.5 GHz, and 17.5 GHz, respectively. As shown in Fig. 2(b), the reflected pulses from four users have time delays of 8.58 ns, 12.49 ns, 15.82 ns, and 19.42 ns, respectively.

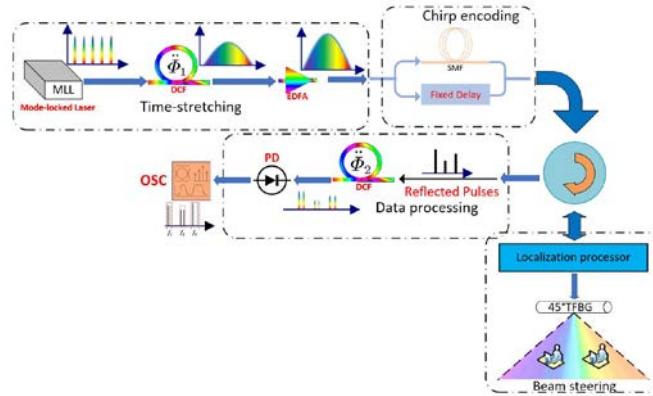


Fig. 1. Schematic showing the proposed indoor user localization system. MLL: mode-locked laser; DCF: Dispersion compensating Fibre; EDFA: Optical Amplifier; SMF: Single-mode fiber; 45° TFG: 45° tilted Fibre Grating; PD: photodetector; OSC: oscilloscope.

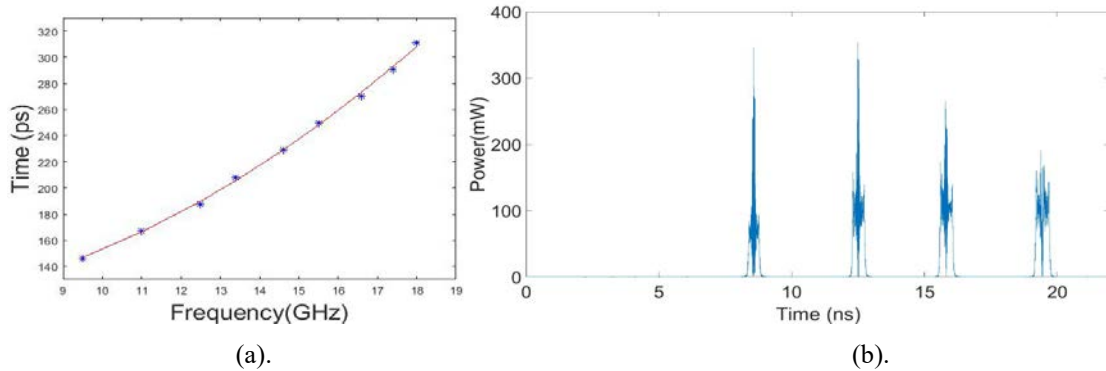


Fig. 2. (a). The measured frequency-time difference mapping (blue-dotted) and its linear fitting result (red line). (b). The reflected optical pulses from different users.

3. Simulation Results and Discussion

Recognizing the central frequency for each reflected pulse is an important step to read the information carried in the pulses. In this paper, Optical FT and STFT are compared to calculate the central frequency separately, shown in Fig. 3 and Fig. 4. STFT can calculate the central frequency precisely (User 1: 11 GHz, User 2: 14.5 GHz, User 3: 15.5 GHz, User 4: 17.3 GHz), shown in Fig. 3. Fig. 4 shows the time delay difference between the two splitted pulses (User 1: 167.05 ps, User 2: 228.85 ps, User 3: 249.45 ps, User 4: 290.3 ps.). Due to the time difference has a linear relation with frequency as Fig. 2(a) shows, so we can get the central frequency for different users: (User 1: 11 GHz, User 2: 14.6 GHz, User 3: 15.5 GHz, and User 4: 17.4 GHz).

The initial wavelength-dependent time delay is determined by the linear relation in the time-stretch process. For these four users, the wavelength-dependent time delay can be calculated (User 1: 5.94 ns, User 2: 7.88 ns, User 3: 8.37 ns, and User 4: 9.39 ns). The actual time-of-flight values for the four users are estimated as 2.64 ns, 4.61 ns, 7.45 ns, and 10.03 ns. Therefore, we can get the absolute distances for different users (User 1: 0.396 m, User 2: 0.69 m, User 3: 1.12 m, and User 4: 1.51 m, respectively).

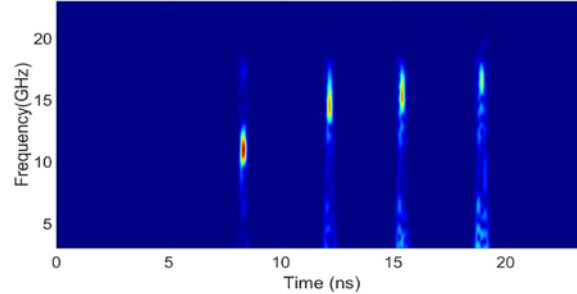


Fig. 3. The spectrum of the reflected pulses.

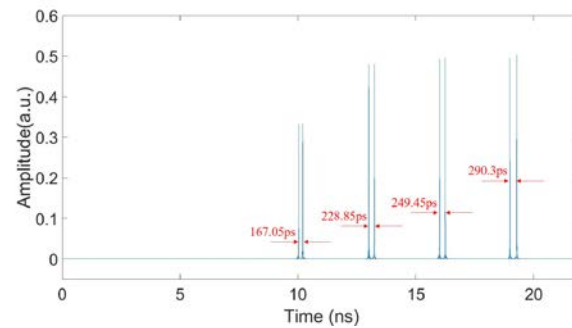


Fig. 4. The output waveform after PD.

4. Conclusions

We have demonstrated a novel approach of ultrafast user localization scheme by using beam steering and optical FT. The reflected pulses from different users are compressed by a dispersion compensation fibre, equal to conduct an optical FT. The result shows that the frequency information carried by pulses can be recognized by observing the time difference on OSC. Moreover, compared with the electronic counterpart, the optical FT can improve signal processing speed. This simulation has verified the feasibility of using optical FT to achieve indoor user localization.

5. References

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